Misbar

An Autonomous AI Platform for Vulnerability Discovery & Risk Assessment

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Introduction

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Misbar is an AI-driven platform that helps organizations discover their attack surface and identify exploitable vulnerabilities on endpoint devices. Aligned with Saudi Arabia's Vision 2030 commitment to cyber resilience, it brings reconnaissance, vulnerability scanning, and AI assistance into a single interactive dashboard. An AI-powered agent handles reconnaissance autonomously by orchestrating multiple industry-standard scanning tools and parsing the results for the user interface, while an active scanning engine probes targets to confirm real, exploitable findings. A live CVE dashboard, an in-platform AI chatbot, and a daily Cyber News Channel round out the platform, turning offensive-security capabilities into a tool that both technical and non-technical users can operate confidently. By unifying these capabilities behind a single, intuitive interface, Misbar lowers the barrier to entry for vulnerability assessment and empowers security teams to act on real-time threat intelligence without requiring deep specialist knowledge.

Problem Statement

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Modern organizations expose a wide attack surface across their endpoint devices, yet identifying which assets are vulnerable, and prioritizing them, remains manual, slow, and dependent on deep security expertise.

Existing security tools operate in silos and produce raw outputs that need skilled analysts to interpret, while AI-assisted tools automate isolated phases without delivering an end-to-end workflow.

Misbar's idea is to help organizations map their attack surface and identify exploitable vulnerabilities on endpoint devices by combining autonomous reconnaissance, active vulnerability scanning, and AI assistance behind a single dashboard.

Aim & Objectives

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Build an AI-driven platform that helps organizations discover their attack surface and identify exploitable vulnerabilities on endpoint devices.

+ Develop an AI-driven agent that runs reconnaissance tools and structures the data.
+ Integrate active vulnerability scanning to identify exploitable findings on endpoint devices.
+ Provide a live CVE dashboard sourcing real-time threat intelligence data.
+ Deliver an embedded AI chatbot to answer questions and explain findings.
+ Operate a daily Cyber News Channel powered by AI-summarized intelligence.

Methodology

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Misbar combines an AI-powered agent for reconnaissance with an active vulnerability scanning engine and a dashboard layer for live CVE awareness and AI assistance.

- **AI Agent: Reconnaissance** *autonomous orchestration of scanning tools*
  The agent runs port scanning, technology fingerprinting, and DNS enumeration tools, manages their execution, and parses each tool's data into a clean, structured form for the dashboard.
- **Vulnerability Scan** *active probing for exploitable findings*
  An active scanning engine probes the target to confirm exploitable vulnerabilities, producing verified findings with CVSS scores, CVE identifiers, and remediation steps.
- **Dashboard: Live CVE Feed** *real-time threat intelligence*
  The dashboard streams the latest CVEs from real-time threat intelligence sources, surfacing live counts, trends, and top affected vendors as they are published.
- **AI Chatbot: Cyber News** *conversational AI & daily intelligence delivery*
  An embedded AI chatbot explains findings in plain language. A Cyber News Channel delivers daily AI-summarized threat intelligence directly to users.

Development Tools

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BACKEND & WEB
- Django (Python)
- HTML5, CSS3, JavaScript
- Supabase (PostgreSQL)
- REST APIs

AI & SCANNING
- DeepSeek v3.2
- OpenVAS
- Reconnaissance Tools
- DNS Enumeration

INTELLIGENCE & LAB
- CVE APIs
- Gemini 2.5 (News)
- WhatsApp API
- Raspberry Pi 5 Lab

System Architecture

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INPUT: Target Endpoint (IP / Domain) ➤ AI AGENT: Reconnaissance (recon & data parsing) ➤ SCANNER: Vulnerability Scan (active probing) ➤ DASHBOARD: Findings + Live CVE (real-time intelligence) ➤ AI CONTEXT: AI Assistant (plain-language Q&A) ➤ NEWS: Cyber News Channel (daily intelligence)

Platform Preview

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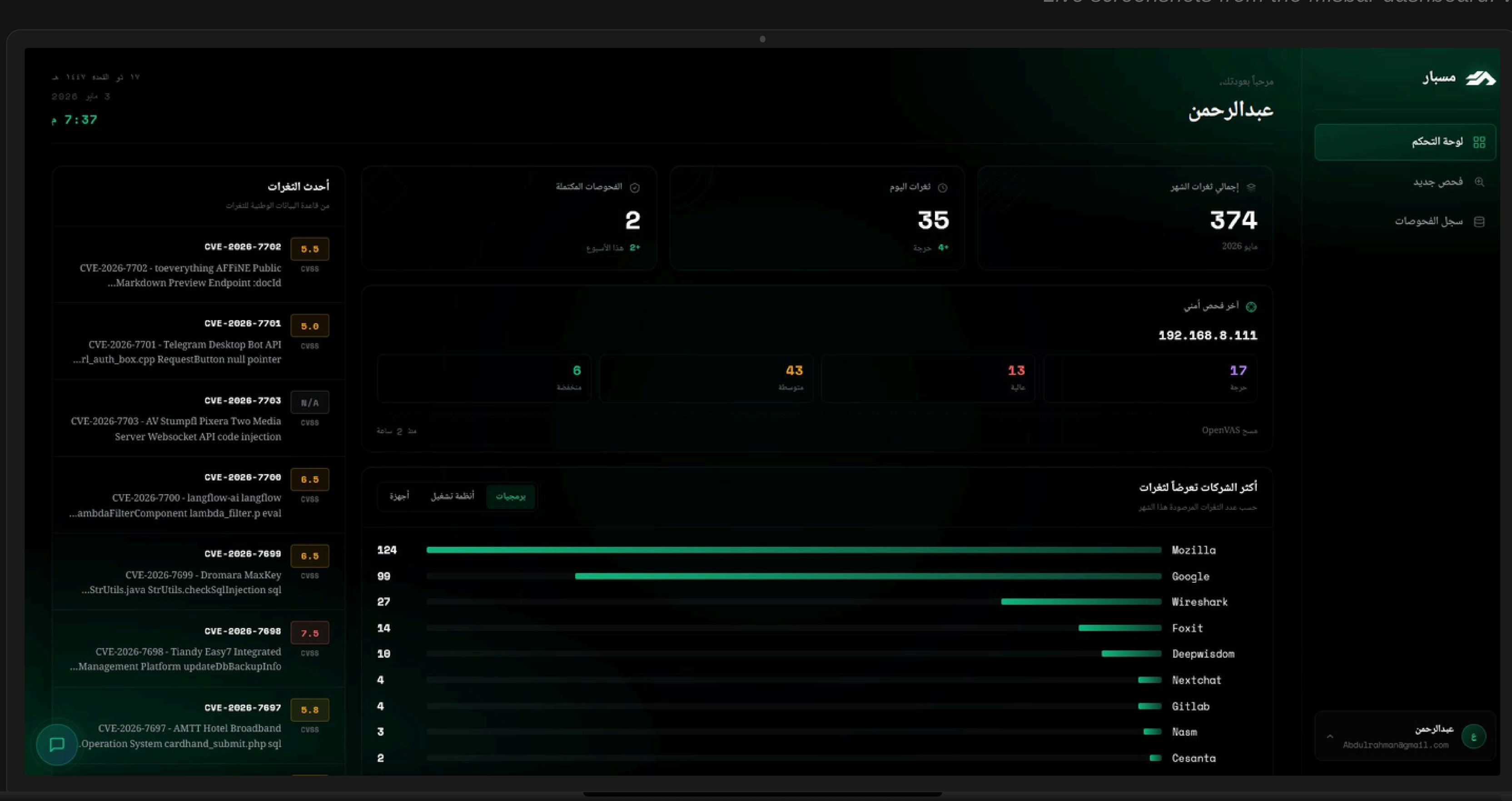
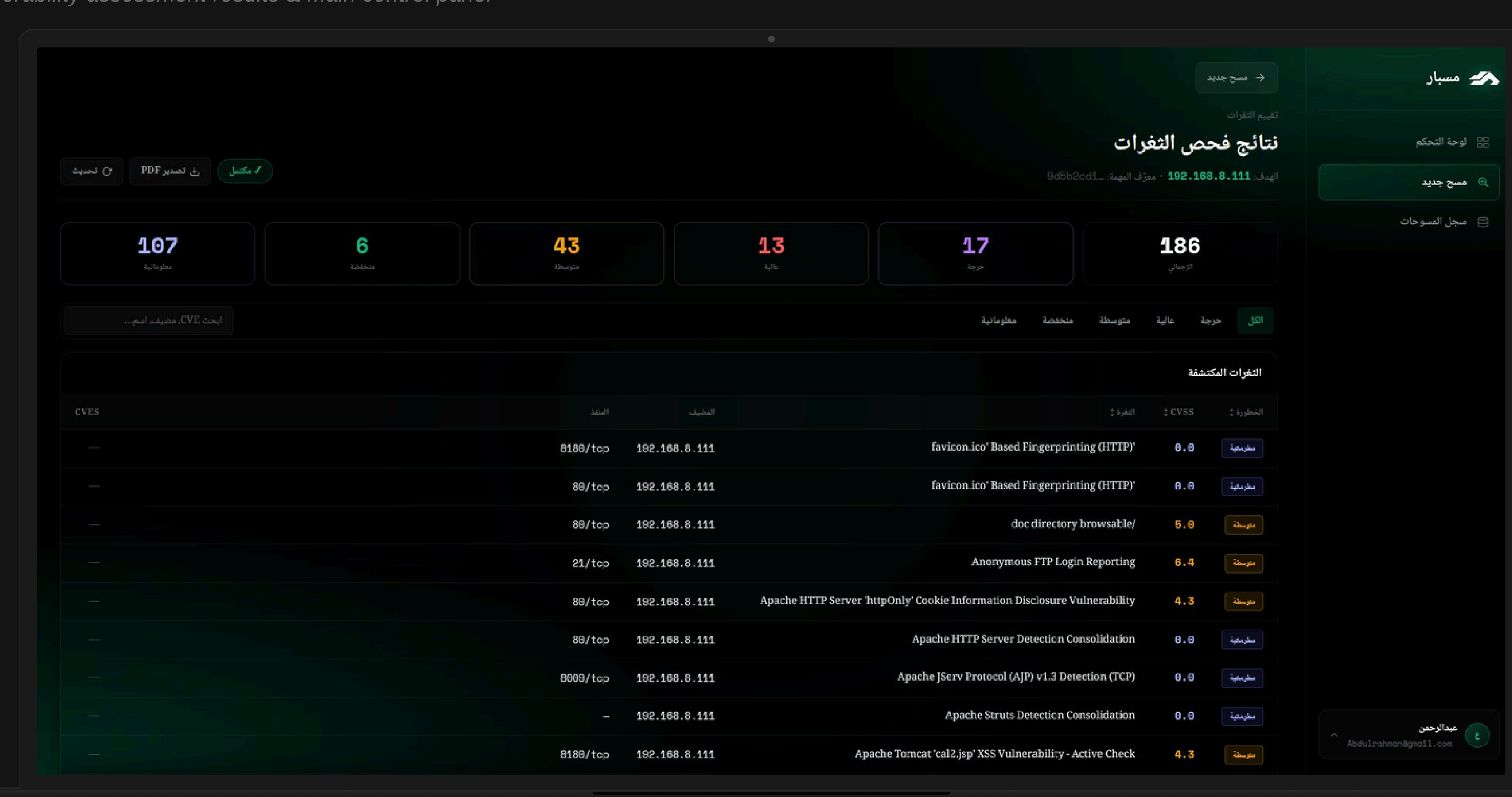
Live screenshots from the Misbar dashboard: vulnerability assessment results & main control panel



Main Dashboard
Live CVE feed · monthly stats · top vulnerable vendors



Vulnerability Scan Results
OpenVAS findings · severity · CVSS · per-host breakdown

Proposed Solution

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Misbar combines an autonomous AI recon agent, an active vulnerability scanning engine, and an interactive dashboard with a built-in chatbot, all behind a single web application.

- AI Recon Agent: an AI-powered agent runs reconnaissance tools and structures the raw output for the dashboard.
- Vulnerability Scan: active vulnerability scan with CVSS scores, CVE identifiers, severity levels, and remediation steps.
- Live CVE Dashboard: real-time CVE statistics including live counts, trends, and top affected vendors.
- AI Chatbot Assistant: embedded chatbot that answers user questions and explains findings in plain language.
- Cyber News Channel: daily threat intelligence summarized by AI, sourced from 100+ cybersecurity feeds.

Conclusion

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Misbar delivers an accessible, AI-driven platform that helps organizations discover their attack surface and identify exploitable vulnerabilities, validated through functional, security, and performance testing on a controlled testbed.

KEY OUTCOMES
- AI-led reconnaissance: an autonomous agent runs and structures the recon data.
- Active vulnerability scanning: confirms exploitable findings on endpoints.
- Live CVE awareness: real-time threat intelligence data.
- Daily Cyber News: AI-summarized threat intelligence channel.

Future work includes additional scanning engines, ticketing system integration, and a fine-tuned threat-intelligence model.

> *Toward accessible, AI-driven cybersecurity.*

PROJECT TEAM

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